Temper

The Future of Banking Infrastructure

Secure mainframe continuity today. The only logical migration path tomorrow.

JavaScript | Python | Java | Rust | C# | **COBOL** | and more to come...

THE REALITY

The world runs on COBOL. That is not changing soon.

95%
of ATM transactions touch COBOL

$3T
in daily commerce processed by COBOL

800B
lines of COBOL in production

Banks do not run COBOL because they love it. They run it because **it works**, it is battle-tested across decades, and the risk of replacing it is existential. Every failed mainframe migration reinforces the same conclusion: **the plumbing stays**.

Source: Reuters, IBM Systems, Micro Focus industry reports

THE CRISIS

The people who understand the plumbing are retiring

The talent cliff

- Average COBOL developer age: **55+**
- Universities stopped teaching COBOL decades ago
- Institutional knowledge leaves when developers retire
- Remaining experts command **$200+/hr** consulting rates

The migration trap

- Rewrite projects average **$100M+** and 5–10 years
- Commonwealth Bank of Australia: **$750M, 5 years**
- Most migrations fail or are abandoned mid-flight
- Failed migrations leave systems **worse than before**

Banks face an impossible choice: maintain systems no one understands, or risk a migration that could bring down critical infrastructure. **Temper offers a third option.**

THE THIRD OPTION

Write in Temper. Compile to COBOL. And to everything else.

Temper Source → `temper build` → **COBOL** | Java | Python | Rust | C# | JavaScript

Today: safer mainframe maintenance

- New business logic written in Temper, compiled to COBOL
- Type-safe, testable code that runs on existing mainframes
- No migration required — the output is native COBOL
- Any developer can write Temper — no COBOL expertise needed

Tomorrow: the migration on-ramp

- Same Temper source compiles to Java, Rust, or C#
- Migrate at your own pace, one module at a time
- COBOL and modern targets run **the same logic**
- Zero behavioral drift between old and new

STRATEGIC POSITION

The only company that can safely bridge both worlds

Temper is uniquely positioned to become the infrastructure layer for banking's next chapter.

CAPABILITY	IBM/MICRO FOCUS	CLOUD REWRITE VENDORS	TEMPER
Maintains existing COBOL	**Yes**	No	**Yes**
Produces new COBOL safely	No	No	**Yes**
Compiles to modern languages	No	Rewrites (risky)	**Yes — same source**
Behavioral equivalence guarantee	N/A	No	**By construction**
Compile-time security guarantees	No	No	**5 defense layers**
Migration risk	Status quo	**Existential**	**Incremental**

No one else can say: **“Use us to maintain your mainframes today, and we are the safest path off them tomorrow.”**

THE MOAT

Every line of Temper a bank writes deepens the relationship

Phase 1: Maintenance

Banks adopt Temper to write new COBOL safely. Modern syntax, type safety, testability. The output runs on their existing mainframes unchanged. **Zero migration risk.**

This solves the talent crisis immediately.

Phase 2: Migration

Every module written in Temper is already portable. When the bank is ready to move off the mainframe, the same source compiles to Java, Rust, or C#. **No rewrite needed.**

Temper becomes the only logical migration vendor.

> *The more Temper a bank writes for COBOL maintenance, the more portable their codebase becomes — and the more locked in they are to Temper as their migration path.*

SECURITY FOR REGULATORS

Compile-time guarantees for regulated infrastructure

Financial regulators increasingly demand proof of secure software practices. Temper provides it structurally.

REGULATORY CONCERN	TRADITIONAL APPROACH	TEMPER APPROACH
SQL injection (CWE-89)	Developer discipline + SAST	**Type system prevents construction**
Mass assignment (CWE-915)	Framework conventions	**Whitelist enforced at compile time**
Cross-platform consistency	Manual testing per platform	**Single source, guaranteed equivalence**
Audit trail	N codebases × N reviews	**1 source + compiler = 1 audit**
Incident response SLA	Weeks (coordinated patches)	**Hours (single rebuild)**

When OCC or Fed examiners ask how you ensure security across platforms, the answer is not process. **The answer is the compiler.**

PROVEN TODAY

A type-safe ORM running in 6 languages with 0 vulnerabilities

We built a production-grade SQL ORM with 5 compile-time defense layers, compiled it to 6 backends, and subjected each to independent MITRE ATT&CK / CWE analysis.

0	6	1
SQL injection vulnerabilities	language backends from single source	commit to fix all findings

An equivalent AI-generated ORM in a single language produced **9 critical/high vulnerabilities** (CVSS 9.8). Temper's type system made every one of those attack vectors **structurally impossible**.

The COBOL backend extends this same guarantee to mainframe-targeted output.

MARKET OPPORTUNITY

Banking's $16B mainframe problem needs a new answer

$16B
annual mainframe modernization spend

72%
of banks say mainframe modernization is critical

Addressable market

- **Top 50 global banks** all run COBOL mainframes
- Insurance, government, and healthcare share the same stack
- Each institution spends **$50M–$500M/yr** on mainframe ops
- Modernization budgets are growing as talent retires

Temper's wedge

- Start with **maintenance** — lowest risk, immediate value
- Expand to **new module development** in Temper
- Grow into **incremental migration** when bank is ready
- Become the **platform** for banking infrastructure

Source: Deloitte Mainframe Modernization Survey, Gartner

THE VISION

Temper becomes the plumbing of modern banking

Banking infrastructure has two requirements: it must be secure, and it must never go down. Temper is built for both.

Near term (12–24 months)

- COBOL backend release
- First banking pilot: new modules in Temper → COBOL
- Regulatory compliance toolkit
- IBM partnership exploration

Long term (3–5 years)

- Temper as the standard for cross-platform banking logic
- Incremental mainframe migration tooling
- Security certification for Temper compiler output
- The platform banks build their future on

> *The company that solves mainframe maintenance becomes the company that owns mainframe migration. Temper is that company.*

WHY NOW

Three forces converging

1. Talent exodus

The COBOL workforce is retiring faster than it can be replaced. Banks need a way to write mainframe code without COBOL expertise. **Temper is that way.**

2. AI code generation

AI generates plausible but insecure code. Without type-level guardrails, AI-written banking logic is a regulatory nightmare. **Temper makes AI output safe by construction.**

3. Regulatory pressure

OCC, Fed, and global regulators are tightening software security requirements. Compile-time guarantees are the strongest evidence a bank can present. **Temper provides that evidence.**

These three forces create an urgent, growing market for exactly what Temper offers: **safe, portable, auditable infrastructure code**.

The opportunity

$16B
annual mainframe modernization market

0
competitors that compile to COBOL + modern

1
source of truth for all platforms

Banks do not need another rewrite vendor. They need a bridge.

Temper is the bridge between the mainframe era and whatever comes next.

Maintain COBOL today. Migrate tomorrow. Same source. Same security. Zero risk.

github.com/temperlang/temper